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20. ABSTRACT *(Continue on reverse side if necessary and identify by block number)*

This Test Operations Procedure (TOP) provides procedures for tropic testing of chemical munitions, weapons, and equipment to determine their ability to withstand the humid tropic environment. It specifies procedures and conditions under which testing, data collection, and data evaluation are to be performed. The TOP is applicable to the basic climatic design type, Quadripartite Standardization Agreement (QSTAG) equivalents B1 and B2, as specified in AR 70-38. The procedures are oriented to tests on chemical offensive weapons, defensive equipment, protective equipment, and decontaminating equipment. The TOP

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Block 20 (cont)

divides tropic testing into two parts--Test Conduct and Test Data. It also treats each of these parts according to initial inspection, functional performance, short-term storage, surveillance (long-term storage), maintenance evaulation, human factors, security from detection, and value analysis. The TOP provides a guide for test planning and should be used in conjunction with other appropriate Test Operation Procedures for specific commodity items undergoing tests.

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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE-RP-702-107
*Test Operations Procedure 8-3-512 15 December 1981
AD No. A108944

TROPIC TESTS OF CHEMICAL EQUIPMENT

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1. SCOPE

a. This Test Operations Procedure (TOP) provides test methodology and techniques for determining the capability of chemical munitions, weapons, and equipment to withstand exposure and function effectively in the humid tropics. A principal reason for tropic testing is to compare data obtained before and after dynamic and static storage tests for assessing, and ultimately predicting, materiel failure. Military toxic chemical agents are not used during tropic testing in the Panama area; only simulants are used.

b. This TOP is a guide to use in conjunction with the appropriate TOPs for the specific commodity item undergoing tests. Reference is made to other TOPs for specific test procedures to be followed in assessing physical condition, technical characteristics, and operating performance of the test item.

*This TOP supersedes TOP 8-1-002, 10 September 1974.

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c. The scope of this TOP specifies applicable procedures under which testing is to be performed, specifies data collection relating test item performance to continued tropic exposure, and evaluates these data to determine the effectiveness of the test item in the tropics.

(1) The material presented is intended for the Basic Climatic Design Type, Constant and Variable High Humidity Daily Cycles, as specified in AR 70-38.[1] Guidance for testing equipment in chambers will not be addressed.

(2) These procedures are limited to tests of offensive equipment, defensive equipment, protective equipment, and decontaminating equipment. Offensive systems designed for use with lethal chemical agents will be tested only with relatively nontoxic simulants. These items require exposure to combinations of weather and terrain conditions which exist in the Republic of Panama.

d. The US Army Tropic Test Center (USATTC) report, Materiel Testing in the Tropics,[2] presents details on environmental characteristics of those areas in Panama available for testing, major problems associated with the areas, a description of fungus-inert and fungus-susceptible materials, the corrosion resistant properties of metals, and the major environmental effects caused by exposure in a tropic climate.

e. Chemical test items may disperse, detect and identify, protect against, neutralize, or decontaminate chemical agents. However, also involved in this category are items employing munitions, smoke, and incendiary agents usually identified with conventional weapons. Accordingly, throughout this TOP, chemical items will be grouped together under the principal function performed:

(1) Offensive Equipment. These items are designed to deliver chemical and riot control agents, smoke, or incendiaries to selected targets. They are an outgrowth of existing weapons delivery systems--similar in physical characteristics, but with a chemical (dry or liquid) payload added. Such items are capable of disseminating bulk, micropulverized powders, or liquid agents over large or small areas. Environmental testing includes exposure and performance testing.

(2) Defensive Equipment. This equipment provides friendly troops with sufficient notice of a chemical agent attack so adequate protective measures can be taken. The alarms and kits are based on atmospheric sensing and chemical/electrical reaction devices. Once an alarm is activated, protective measures include collective protection, providing shelter to many individuals who otherwise would be hampered by wearing protective masks. This equipment

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[1] AR 70-38, Research, Development, Test, and Evaluation of Materiel for Extreme Climatic Conditions, 1 August 1979.

[2] USATTC Report No. 700401, Materiel Testing in the Tropics, April 1979, TECOM Project No. 9-CO-150-000-099.

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provides and maintains filtered air under positive pressure at a level precluding the infiltration of hazardous levels of toxic agents. It is adaptable for use with combat vehicles, field medical facilities and other shelters, and certain combat vehicles and vans. Also in this category are items capable of sampling and identifying chemical agents, detecting personnel by chemical means, or marking and tracking personnel. Environmental effects would involve chemical degradation, microbial degradation, and propagation of airborne chemical contamination within the jungle environment.

(3) Protective Equipment. As used here, this term addresses individual protective equipment and systems, such as masks and self-contained environmental suits. These items protect the individual from chemical agent effects by preventing entry of such agents into the respiratory system or by decreasing cutaneous and percutaneous hazards. Testing individual protective equipment involves human engineering aspects which are of paramount concern. Testing protective masks, for instance, involves not only determining if they will deliver chemical-agent-free air to the wearer, but evaluating: (a) comfort of fit while wearing, (b) ease of breathing (inhalation and exhalation), (c) visibility (with and without optical inserts), (d) interference with other equipment carried or used, (e) interference with normal duties, (f) ease of donning, (g) comfort during exposure to high temperature and humidity, (h) fit of the nose-cup, (i) adequacy of speech transmission, and (j) mask stability. Data on technical characteristics are gathered from laboratory and field testing; human factors relationships are assessed by objective measurement techniques developed for many aspects, and by subjective, structured interviews of subjects who participated in field wearing tests. Questionnaire development and administration are performed by a human factors specialist. Individual protective equipment, such as an overgarment, also must be subjected to a human factors evaluation. It must be comfortable to wear, not cause skin irritation, produce a minimum of heat stress, and be easily donned and doffed. Environmental effects would be similar to protection systems, but human mobility and combat performance also may be affected by the tropic environment.

(4) Decontamination Equipment. Decontamination devices provide access to areas to recover personnel and restore equipment. These devices also aid the process of making any person, object, or area safe by removing or neutralizing chemical agents. Caution must be exercised to avoid irritation to eyes, skin, and respiratory tract that is caused by the reaction (toxic fumes or flashing) of some decontamination agents with certain chemicals. If doubt exists concerning the compatibility of chemicals used for decontamination, testing should not begin until qualified personnel are consulted. Environmental effects include chemical degradation of the decontaminating and impregnating chemicals or false reaction of these chemicals with the jungle chemistry. Jungle mobility tests for large decontaminating equipment include functional performance testing, because individual decontamination and reimpregnation kits may be affected by a change in body chemistry produced by the tropic environment.

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f. Test sites used must be isolated to prevent inadvertent exposure to chemical agent simulants, and must be of adequate dimension to allow proper evaluation of dispersion characteristics of a munitions or dissemination device. Terrain, vegetation, surface effects, and prevailing atmospheric conditions are factors to be considered in test site selections. Because of required environmental impact evaluation of testing in Panama, only approved simulant agents can be tested in the open air.

2. FACILITIES AND INSTRUMENTATION

2.1 Facilities. The following test sites and USATTC facilities are available in Panama for evaluating chemical equipment in the tropics:

a. Chemical laboratory.

b. Materials laboratory.

c. Instrumentation laboratory.

d. Human Factors laboratory.

e. Fort Clayton General Purpose Test Area.

f. Fort Clayton POL Tank Farm.

g. Gamboa Forest Human Factors Test Area.

h. Various material exposure sites.

i. Training areas and firing ranges.

2.2 Instrumentation. One or more of the following may be required during the various evaluations:

a. Measuring/recording devices and other equipment peculiar to the test item.

b. Control items, as required.

c. Measuring/recording devices and other equipment listed in the applicable TOP.

d. Chemical agent simulant, riot control agent, and associated containers and dispersers.

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3. PREPARATION FOR TEST

3.1 Test Concepts

a. Test concepts for the Basic Climatic Design Type, Constant and Variable High Humidity Daily Cycles, and functional tests will be performed on a selected basis to determine if the test item meets established criteria.

b. The following test concepts apply when evaluating performance of offensive, defensive, protective, and decontamination equipment items. These concepts are found in TOPs 8-2-082,[3] 8-2-070,[4] 8-2-061,[5] and 8-2-110.[6]

(1) Receipt Inspection (TOP 8-2-500[7]).

(2) Safety.

(3) Human Factors.

c. Special test concepts applicable to certain classes of chemical items are:

(1) Offensive Equipment. For offensive equipment covered in TOPs, the following TOPs provide test methods for the specific storage or performance aspects shown: TOPs 8-2-082, 8-2-093,[8] and 8-2-162.[9]

(a) Radiography (TOP 8-2-509[10]).

(b) Leak Resistance (TOP 8-2-512[11]).

(c) Dissemination Characteristics (TOP 8-2-513[12]).

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[3] TOP 8-2-082, Dispersers, Riot Control Agent, Portable, 2 October 1967.
[4] TOP 8-2-070, Chemical Agent Detector Kits, 31 October 1967.
[5] TOP 8-2-061, Decontaminating Apparatus, Portable, 30 September 1967.
[6] TOP 8-2-110, Masks, Protective, October 1980.
[7] TOP 8-2-500, Receipt Inspection, 30 December 1967.
[8] TOP 8-2-093, Hand Grenades, Riot Control, 31 October 1967.
[9] TOP 8-2-162, Warheads, Rocket, Chemical Agent, 17 February 1968.
[10] ~~TOP 8-2-509, Radiography, 31 January 1968.~~
[11] TOP 8-2-512, Leak Testing of Chemical Agent-Filled Munitions and Containers, 1 November 1971.
[12] TOP 8-2-513, Dissemination Characteristics, Chemical Munitions/Dissemination Devices, 1 November 1971.

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(2) Defensive Equipment. For defensive equipment covered in TOPs, the following TOPs provide test methods for the specific storage or performance aspects shown: TOPs 8-2-070, 8-2-072,[13] 8-2-191,[14] and 8-2-193.[15]

(a) Field Detection.

(b) Emergency Adaptability.

(c) Alarm Testing.

(3) Protective Equipment. For protective equipment covered in TOPs, TOP 10-2-021[16] provides test methods for the specific storage or performance aspects.

(4) Decontamination Equipment. For decontamination equipment covered in TOPs, the following TOPs provide test methods for the specific storage or performance aspects shown: TOPs 8-2-061 and 8-2-063.[17]

(a) Leak Resistance (TOP 8-2-511[18]).

(b) Assembly and Disassembly Aspects.

(c) Medical Examination.

3.2 Test Planning

a. Test plans outline the approach, test objectives, test criteria, methods, scope, and test length, including those subtests necessary to determine the degree to which a test item meets the prescribed criteria. A test plan should be complete in itself, with reference to applicable TOPs and required manuals.

b. Upon receipt of the test directive, the plan is prepared based on a study of the established criteria, a literature search, discussions with experienced individuals (e.g., chemical, microbiological, materials, human factors, statistical, and meteorological specialists), and numerous planning conferences. At the detailed planning stage, environmental tests are designed specifically to insure that reliability requirements can be evaluated on the basis of resulting data.

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[13] TOP 8-2-072, Sampling and Analyzing Kits, CBR Agent, 3 March 1970.
[14] TOP 8-2-191, Alarms, Chemical, 27 October 1967.
[15] TOP 8-2-193, Collective Protection Systems, Field Shelters, 30 November 1967.
[16] TOP 10-2-021, Combat Uniforms and Protective Equipment, 6 February 1973.
[17] TOP 8-2-063, Decontaminating Kits, Individual, Field, 10 December 1969.
[18] TOP 8-2-511, Leak Testing of Protective Equipment, 29 February 1968.

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c. In preparing for a test, the test officer and other test personnel must:

(1) Conduct a thorough study of requirements stated in the requirement documents, TOPs, and test directive to insure that test criteria are understood.

(2) AR 70-10,[19] paragraph 2-5c and 2-16a requires evaluation of the environmental impacts identified in the National Environmental Protection Agency documentation prepared by the sponsor for the item/system to be tested and/or identification of effluents or emissions from equipment required to meet specifications of the law. HQ, TECOM will identify in Independent Evaluation Plan/Test Design Plans or otherwise specify those subtests that Field Installation/Field Operating Activity must perform to meet this requirement.

(3) Review instructional material issued by the developer or manufacturer and reports of previous tests conducted on similar equipment.

(4) Prepare a detailed test schedule showing proposed time periods allotted for each subtest. Test periods will depend on test type [e.g., short-term storage, surveillance (long-term storage), functional, or developmental test].

(5) Prepare record forms and test logs for systematic entry of data, chronology of testing, and evaluational analysis of the test item's suitability for use in the tropics. Review the safety release (a TECOM safety release must be received before testing begins). Tabulate the safety precautions that must be followed to insure safe operation of the test item, and include all applicable safety manuals and instructions.

(6) Verify that test facilities, equipment, and accessories are available, operational, and properly calibrated for the appropriate time limit (e.g., power sources, when used, will be checked to insure correct voltage outputs).

(7) Make final arrangements for test support, including participating agencies, activities, and facilities. Test support includes meteorological, topographical, human factors, statistical, chemical, microbial, and materials analyses.

(8) Obtain and have available all operating instructions for test instruments to be used during testing.

(9) Insure that all historical/technical data and specifications are available to test support personnel during testing.

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[19] AR 70-10, Test and Evaluation During Development and Acquisition of Materiel, 29 August 1975.

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(10) Plan the use of photooptical techniques, where necessary, to record and document test results. Video tape recordings also are useful for evaluating test item and test personnel performance.

d. Test personnel training and familiarization procedures apply to the test item and the control item, when used. The test officer and other designated personnel must evaluate the adequacy of all draft technical manuals and safety instructions used by test personnel. This evaluation continues throughout the course of the test. Instructions are given in the following table:

TABLE 1. TYPES OF INSTRUCTION FOR CLASSES OF PERSONNEL

Personnel	Types of Instruction
All	Purpose of the test. Characteristics of test item. Characteristics of control item, if any. Decontamination procedures. Safety aspects of the test item. Terrain in which testing is to be conducted. Kind of data to be obtained.
Test Item and Control Item Operators	Test item (and control item, when used) operating characteristics and limitations. Expected test item performance. Safety precautions. Decontamination procedures. Kind and extent of all maintenance actions to be taken under tropic environmental conditions. Operating procedures followed under tropic environmental and terrain conditions. Procedures to follow and purpose of filling out checklists and questionnaires.
Test Evaluators/Human Factors Personnel	Purpose and use of all checklists and questionnaires, and methods used in their evaluation. Calculations for evaluating test data.
Topographical Analysis Support Personnel	Physical terrain features to be measured (e.g., stream water velocity and terrain slopes). Vegetation features to be noted and described. Materiel inspections and analyses to be performed.

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Table 1 (concluded)

Personnel	Types of Instruction
Maintenance Support Personnel	Scheduled maintenance requirements. Procedures, equipment, and materials to be used for emergency repairs and unscheduled maintenance. Maintenance records to be completed.
Meteorological Support Personnel	Terrain areas which will be used. Calendar dates of all tests. Measurements to be taken.
Storage and Surveillance Personnel	Storage, handling, layout, spacing, pest control, fire protection, security, and inspection procedures.
Storage and Surveillance Personnel (cont)	Organizational maintenance to be performed--including cleaning and microbial and materials inspections. Calendar dates of all inspections.
Medical Assistance Personnel	Type of vehicle required. Dates required. Special equipment and supplies required.
Explosive Ordnance Disposal (EOD) Personnel	Type and class of munitions under test. Terrain areas to be used. Decontamination procedures. Disposal requirements and procedures.
Chemical Safety Officer	Characteristics of test and control items. Safety release obtained before testing begins. Dates of operational use. Decontamination procedures. EOD and medical assistance contacts. Communication assistance required during test conduct.

4. TEST DESIGN/ANALYTICAL PLAN

a. The reliability characteristics set forth in the applicable criteria are usually stated as "best operational capability" (BOC) and as a "minimum acceptable value" (MAV). Tests required to demonstrate achievement of the

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BOC and MAV will have been addressed in Section IV [Coordinated Test Program (CTP)] of the Development Plan (DP). Statistical confidence levels and risks associated with demonstrating achievement of reliability characteristics also will be included in the DP. If confidence levels are not specified in the DP, a point estimate and 80- and 90-percent, two-sided confidence intervals of reliability should be computed.

b. If the failure distribution can be assumed to be exponential, MIL-STD-781C[20] should be used to determine test time and sample size. Reliability analysis using the exponential distribution is outlined in paragraph F-5 of US Army Materiel Command (AMC) Pamphlet 702-3.[21]

c. If the binomial distribution is used, the sample size should be large enough to demonstrate achievement of the minimum acceptable value at 90-percent confidence using the lower bound (unless stated otherwise in the DP). The binomial distribution is used extensively for "one-shot" items such as grenades, rockets, and land mines.

d. When a one-shot device is being tested, the items will be divided into equal groups and a cyclic schedule designated. This cycle should encompass the various changes in meteorological conditions.

e. Before any exposure cycling, it is imperative to determine if the test item should be exposed first to the dry season or to the wet season. It is possible that different results will be obtained if the item is exposed first to the dry season than if it is exposed first to the wet season. The criterion to select one type of season depends on the nature of the performance test, materiel tested, materials involved, storage mode, and site, as shown in table 2.

TABLE 2. CYCLIC TEST SCHEDULE

Cycle	Type of Test to be Performed
No. 1	Initial performance (including control sample)
No. 2 (3-month)	Dry season/wet season performance
No. 3 (6-month)	Short-term storage performance
No. 4 (9-month)	Wet season/dry season performance
No. 5 (12-month or XX-month)	End of test performance (including control sample)/or long-term storage (surveillance)

[20] MIL-STD-781C, Reliability Tests Exponential Distribution, 21 October 1977.
[21] AMC Pamphlet 702-3, Reliability Handbook, 28 October 1968.

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When a long-wearing device (e.g., a portable decontamination apparatus) is being tested and can be repaired many times without an appreciable change in performance characteristics, only one group of test items from the same production lot is required, provided that the item is restored to an "as-new" condition at the start of each of the environmental tests described above. A similar approach can be used without significant loss in accuracy when testing medium-wear devices (e.g., protective clothing or shoes).

f. If not otherwise specified in the DP, a statistical plan for the major chemical groups can be established as follows:

(1) Offensive equipment--use procedures in paragraph 6.1 of TOP 3-4-003,[22] including division of the lot into five test groups.

(2) Protective equipment--use procedures in paragraph 6.1.3.2 of TOP 3-4-003, except substitute the word "wear" in place of "fire," and the word "equipment" in place of "weapon."

(3) Defensive and decontamination equipment--use procedures in paragraph 6.1.3.2 of TOP 3-4-003, except substitute the word "operate" in place of "fire," and the word "device" in place of "weapon."

g. When a control item is used for comparison purposes, follow the same test procedures as with the test item.

h. In lieu of a discussion on the choice of distributions other than binomial or exponential, refer to the following: AMC pamphlet 702-3; TOP 3-1-005;[23] and National Bureau of Standards (NBS) Handbook 91.[24]

5. PERFORMANCE TESTS

5.1 Initial Inspection. Determines the condition of the test item upon arrival at the installation site, and assesses its characteristics before exposure for any appreciable length of time to the tropic environment in which it will be used. Refer to TOP 8-2-500 for detailed receipt inspection of chemical items. Immediately upon receipt of the test item:

a. Inspect and record the condition and identification markings of all packages and packing, and conformance to MIL-STD-794D[25] for overseas shipment (Level A or Level B, as applicable).

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[22] TOP 3-4-003, Armament and Individual Weapons, 28 January 1971.
[23] TOP 3-1-005, Field Artillery Statistics, 1 March 1972, with Change 1, 10 June 1974.
[24] NBS Handbook 91, Experimental Statistics, October 1966.
[25] MIL-STD-794D, Procedures for Packaging and Packing of Parts and Equipment, 15 December 1972.

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b. Weigh and measure the individual package for each packaged item component, or a sample package if identical components are packaged identically. Unless otherwise specified, the minimum sample size will conform to guidelines in table 3 when variability is small in relation to item requirements. If necessary, refer to MIL-STD-414[26] for addditional sampling procedures.

TABLE 3. SAMPLE SIZE FOR STATISTICAL RELEVANCE

Number of Items	Minimum Sample Size	Number of Items	Minimum Sample Size
1 to 4	All	10	6
5	4	11 to 15	7
6	4	16 to 40	8
7	5	41 to 80	9
8	5	81 to 1000	10
9	6	1001 or more	1%

c. Unpack and remove temporary preservative coating on test item, if applicable.

d. Verify completeness of the test item, associated components, and system support package, as specified in the Basic Issue Item List (BIIL), and submit an Equipment Performance Report (EPR) for any damages or omissions (AMC Regulation 700-38[27]).

e. Inspect all surfaces for proper protective and identification painting, and other protective coatings. Take color photographs of all surfaces displaying roughness, corrosion, or microbial indicators; when possible, include a calibrated color patch standard. If corrosion or microbial growth is suspected, submit item for laboratory inspection and analysis.

f. Conduct a pretest inspection in accordance with the item's draft technical manual, TOP 8-2-500, and TOP 8-2-512, as applicable, and repair or replace all test items or components not passing this inspection.

5.2 Functional Performance. Evaluates the capability of the test item to perform its intended mission in the tropic environment.

a. Perform testing during both the dry season and wet season--normally January to mid-April and mid-April through December, respectively.

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[26] MIL-STD-414, Sampling Procedure and Table for Inspection by Variables for Percent Defective, 11 June 1957, with Notice 1, 8 May 1968.

[27] AMC Regulation 700-38, Test and Evaluation--Incident Disclosed During Materiel Testing, 25 October 1974.

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b. Perform functional tests for a sufficient number of missions to obtain a suitable measure of the test item's probability of mission success. When specified, these tests will include subjecting the test item to shock, vibration, weather, water immersion, and vegetation conditions encountered during cross-country, secondary road, or airborne operations, as applicable. Refer to Field Manual (FM) 3-8[28] and FM 21-40[29] for chemical/biological (CB) field operations information.

c. During operations, verify that no terrain conditions or obstacles exceed the limits allowed for the test course.

5.3 Storage. When the number of test items permits, a control group sample will be stored in an air-conditioned area not influenced by tropic environmental effects. Experimental storage samples will be compared with control samples for tropic storage causal analyses.

a. Short-Term Storage. Determines any deterioration in the test item during and after short-term shorage. Short-term storage is classified as a storage period of less than 12 months.

(1) Store the test item or components consistent with normal practice for the commodity class of the test item, as supplemented by the applicable information in the draft technical manual.

(2) Place the test item or components in the area designated for short-term storage. Unless otherwise specified, make minimum use of preservatives.

(3) Periodically, inspect the test item for evidence of corrosion, microbial growth, or other changes in physical appearance which may be caused by tropic environmental exposure. Submit the test item displaying environmental effects to technical personnel for analysis and evaluation. Unless otherwise specified, make these tests only on a sample quantity of test items as specified in table 3, above. When chemical items are being tested, at least one sample container should be disassembled by technical personnel during an inspection to determine if there has been any chemical reaction during the storage interval. A technical evaluation should be made to determine if excessive amounts of moisture have penetrated the chemical containers, or a leak test should be performed.

(4) At completion of the storage period, repeat functional testing procedures to determine if any deleterious effects have been induced in the test item by storage.

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[28] FM 3-8, Chemical Reference Handbook, 6 January 1967, with Changes 1-5.
[29] FM 21-40, NBC (Nuclear, Biological, Chemical) Defense, 26 August 1977.

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b. Long-Term Storage (Surveillance). Determines any deterioration in the test item during and after long-term storage (long-term storage is 12 months or longer) when properly packed and packaged for storage in a tropic environment.

(1) Perform inspections listed above for short-term storage; additionally, perform the following:

(a) Unless otherwise specified, store all items in their normal long-term storage containers, as specified in the draft technical manual.

(b) Determine the numbers and types of inspections/tests to be conducted during the storage period. Unless otherwise specified, plan for five cycles of tests/inspections equally spaced throughout the storage period (table 2, above).

(c) Determine at which "cycle number" dates (month, day, and hour) the specified number of test items are to be tested or inspected. Unless otherwise specified, test not fewer than the minimum sample size (determined from table 3, above) for each specified cycle number date.

(d) Store the test items in the specified environment, using the applicable procedure described in Technical Manual (TM) 743-200,[30] unless otherwise specified. Note the particular month, day, and hour that this storage takes place as the "cycle No. 1" date.

(2) At each cycle number date, perform the following on each test item in the test group:

(a) Visually examine the item and take color photographs of all corrosion and fungal growth areas. These environmental effects should be evaluated by technical personnel.

(b) Analyze all materials which display evidence of microbial attack, as specified in TOP 8-2-514.[31]

(c) Accomplish all specified operator maintenance and repairs.

(d) When a functional test is performed, transport the specified number of test items to the test site, perform the test required, and determine the extent to which the performance characteristics of the test items have deteriorated during the storage period.

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[30] TM 743-200, Storage and Materiel Handling, 28 June 1955, with Changes 1-27.
[31] TOP 8-2-514, Microbiological Air Sampling in the Tropics, 28 March 1972.

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(3) During tropic tests in Panama, when time constraints limit storage tests to fewer than 12 months, obtain complete meteorological measurements. However, for long-term storage periods (12 months or longer), historical meteorological data can be used because meteorological parameters (relative humidity, temperature, and pressure) can be defined adequately for the purpose of the test by historical measurements.

5.4 Logistics Supportability. Evaluates maintenance characteristics of the test item and suitability of the systems support package in a tropic environment. Data expressing system maintenance factors are collected in preparing for and completing active testing. If, at the end of testing, very little maintenance has been performed, representative samples of the maintenance procedures in the draft technical manual should be completed to determine the ease in performing these actions, with emphasis on the following:

a. Organizational (O), direct support (F), and general support (H) maintenance requirements.

b. Operator through general support maintenance literature.

c. Repair parts.

d. Tools.

e. Test measurement and diagnostic equipment.

f. Calibration and maintenance facilities.

g. Personnel skill requirements.

5.5 Safety. Evaluates unsafe item characteristics and possible hazards to which personnel are exposed.

a. Before the test begins, the test officer will insure that a safety assessment report or safety release has been received and is understood. The safety documentation will include information on functional limitations and hazards peculiar to the test item.

b. Safety procedures during the test will conform to DARCOM Regulation 385-12[32] to enable TECOM evaluation. The cardinal principle is to limit exposure to a minimum of personnel for a minimum time, to a minimum amount of hazardous material consistent with efficient operations. Technical information on hazards and safety characteristics, as provided by the safety statement and other pertinent information, will be considered. Such information

[32] DARCOM Regulation 385-12, Life Cycle Verification of Materiel Safety, 20 September 1976.

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will consist of evaluating potential hazards, risk analysis, limitations, and precautions (including special test equipment and techniques incorporated in testing procedures).

c. A specific individual (chemical safety officer) will be charged with responsibility for safety. He will be familiar with construction and operation of the test item and its critical components, will have full knowledge of hazards and safety aspects of the test, and will review test procedures to evaluate potential hazards and recommend control measures.

d. To insure proper test procedures, all personnel participating in or observing the tests will be briefed on hazards involved and precautions required.

e. Test personnel will observe all safety precautions specified in AMC Regulation 385-100,[33] US Army Materiel Development and Readiness Command (DARCOM) Supplement 1 to AR 385-10,[34] and the draft technical manual.

f. Protective masks and appropriate protective clothing will be worn by test personnel when handling riot control and chemical simulant filled munitions and containers.

g. EOD personnel will dispose of any test ammunition or explosive considered unsafe.

h. Test personnel will observe proper safety precautions during testing, record conditions that might present a safety hazard, the cause of such hazard, and steps taken to eliminate the hazard.

i. After testing, if the item is determined safe for use, the test officer will include a safety evaluation in the final report. If the item is found to be unsafe, the test officer will provide a detailed explanation in the final report.

5.6 Human Factors (HF). Evaluates those soldier-related design and performance characteristics that degrade system effectiveness in the tropic environment.

a. Coordinate with HF personnel to develop task/item checklists and questionnaires reflecting HF design considerations. Checklists will allow supervisory test personnel to compare item features against appropriate criteria and record comments to evaluate suitability of the man-item interface, with particular emphasis on operations under adverse weather conditions.

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[33] AMC Regulation 385-100, AMC Safety Manual, 21 April 1970, with Change 1.
[34] DARCOM Supplement 1 to AR 385-10, Army Safety Program, 1 February 1980.

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Detailed criteria and HF considerations for each task may be found in TECOM Supplement 1 to AR 602-1,[35] Human Factors Evaluation Data for General Equipment (HEDGE) Test Procedures (Part I),[36] HEDGE (Part II),[37] and MIL-STD-1472B.[38]

b. During or after performance testing, HF personnel will conduct objective tests and subjective interviews with test personnel to determine soldier performance time and errors, and their impact on system effectiveness in the humid tropics.

5.7 Laboratory Analysis. Includes chemical, microbial, and materials deterioration, as required.

a. Basic chemical and materials test data, if applicable, will be furnished by laboratory personnel.

b. Laboratory analysis will be conducted on the test item and the control item for comparison purposes.

c. Laboratory data must be obtained during initial inspection for baseline values, and periodically during the test to assess materiel item failures and tropic effects.

5.8 Security from Detection. Evaluates capability of the test item to avoid detection when stored or functionally tested in the humid tropics.

a. Position test item in designated test site.

b. Use indigenous material, nets, tarpaulins, and any other prescribed camouflage material to conceal test item. Place camouflage material about the site to enhance concealment qualities.

c. Observe the concealed item by ground and aerial means at various times of the day and at varied distances from the test item.

5.9 Value Analysis. Analyzes the primary function and features of the test item for cost reduction purposes without compromising performance, reliability, quality, maintainability, or safety.

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[35] TECOM Supplement 1 to AR 602-1, Human Factors Engineering (HFE) Program, 1 June 1977, with Change 1, 8 December 1978.
[36] Human Factors Evaluation Data for General Equipment (HEDGE) Test Procedures (Part I), 20 December 1977.
[37] HEDGE (Part II), 20 December 1977.
[38] MIL-STD-1472B, Human Engineering Design Criteria for Military Systems, Equipment and Facilities, 31 December 1974.

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a. During conduct of all tests, test personnel will evaluate the test item from a value-versus-cost standpoint, and record pertinent comments concerning features or components which can be eliminated or modified to accomplish cost reduction without impairment of performance, reliability, quality, maintainability, or safety. The applicable portions of DARCOM Regulation 70-8[39] will be used for evaluation.

b. Consideration will be given to the topics listed below, with appropriate comments for each topic.

(1) Mission capacity. The test item should be capable of accomplishing the specified task with only a reasonable margin of excess capability. Excess capacity and unused capability normally result in unnecessary bulk, excessive weight, and unwarranted costs.

(2) Simplicity. Unnecessarily complex components and systems, redundancy, and unneeded parts will increase costs and maintenance efforts.

(3) Level of technology. In many instances, the use of recently developed, currently available components and automated features will result in an overall product improvement and cost savings.

(4) Standardization. The use of identical parts, and parts currently in the military system, will reduce the overall logistics burden.

(5) Materials and methods of construction. Polished surfaces, overdone finishes, and expensive materials will result in unnecessary costs if used indiscriminately and unnecessarily.

(6) Tolerances. Inadequate tolerances will result in difficulties and delays in accomplishing assembly, routine maintenance, and repair.

6. DATA REDUCTION AND PRESENTATION

6.1 Test Data

a. General Requirements:

(1) Record each measurement and the arithmetic mean of the measurement. Analyzing criteria by the attributes approach, which is subject to small deviations, requires a statistical sampling of measurements under identical test conditions. This does not mean that statistical sampling of measurements should be limited to the attributes approach. Also, application of statistical methods are probably more important when measurements are subject to large deviation. In any event, the size of measurement errors should not limit the use of statistical methods.

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[39] DARCOM Regulation 70-8, Value Engineering, 23 February 1978, with Change 1.

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(2) Indicate the accuracy of the measuring device employed, and record its calibration date, when applicable.

(3) Describe or photograph progressive degradation observed on any part, and show the "before" and "after" condition when recording. Submit the particular item for technical evaluation and analysis.

(4) Show terrain profile, vegetation, and density descriptions in the operations log, if applicable.

(5) Show, in tabular form, the scoring values used, and scoring results obtained, when personnel are asked to fill out a questionnaire giving opinions on specific features in a particular test. When an analysis of variance or another statistical test is used to test for differences in scores, include the results of this analysis.

(6) Record the following, and any other significant details, in the test log or on record forms:

(a) Actual test schedule.

(b) Month, day, and hour of start and finish of short- and long-term (surveillance) storage periods.

(c) Types, lengths, and dates of cross-country test courses used.

(d) Exceptions encountered to the safety release.

(e) All safety precautions followed.

(f) Exceptions encountered to the environmental impact assessment.

(g) Test facilities, equipment, accessories, and operating instructions used.

(h) Supporting or participating agencies.

(i) Photographic documentation (what, where, negative number).

(7) Personnel Training and Familiarization. Record the following:

(a) MOS.

(b) Rank.

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(c) Unit.

(d) Experience.

(e) Previous training and skill level.

(f) Adequacy of draft technical manual for support personnel training.

(g) Requirements for training aids.

(h) Difficulties encountered in completing training program.

(8) Statistical Plan. Record the following:

(a) The statistical method to be used in determining probability of mission success.

(b) Number of identical missions to be made.

(c) Control item to be used for comparison purposes.

(d) Personnel to be used in the various tests.

(e) Research design to be followed, including provision for control group.

(f) Dependent and independent variables and units of measurement.

(g) Statistical tests to be used in analyzing data.

b. Initial Inspection. This inspection should include the areas listed below, with appropriate records.

(1) Conformance with specified packing, packaging, preserving, identifying, and marking criteria.

(2) Any omissions of components or equipment compared to shipping list, damages detected, or repairs made.

(3) All physical characteristics of the test item, as specified in the applicable commodity item TOP and TOP 8-3-500.

(4) Missing or improper protective coatings on the test item, and photographs of roughness, corrosion, or microbial indicators.

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c. Functional Performance. Immediately after assembling and positioning the test item in the physical environment in which it will be used, perform the following:

(1) Record performance characteristics in all specified modes and ranges under the applicable meteorological conditions.

(2) Describe scope and test procedures used for evaluating performance of the test item.

(3) Provide a sketch map (or photomap) of the test area and a description of the environment.

(4) Furnish meteorological data for each test, where applicable. In some instances (e.g., long-term storage), historical data may describe the environmental conditions sufficiently.

d. Storage (Short-Term Surveillance). Provide descriptions and records of the following:

(1) Condition of the test item just prior to storage, and storage preparations, including (if applicable) containers and packaging used.

(2) Storage environment and daily meteorological conditions.

(3) Number of test items involved.

(4) Storage method and duration for each item.

(5) Calendar dates and results of all inspections.

(6) Maintenance and repairs performed.

(7) Extent of deterioration of operating characteristics.

e. Logistics Supportability.

(1) Scheduled maintenance. Record the following:

(a) Description of test phase, environment, and meteorological factors associated with maintenance performed.

(b) Procedures and special tools (if any) required to complete maintenance actions; description of maintenance performed.

(c) Component or feature involved.

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(d) Identify of, and elapsed time since, previous maintenance (scheduled or unscheduled, whichever is shorter).

(e) Parts required, if any.

(f) Time required to complete maintenance actions.

(g) Any difficulty in completing maintenance actions.

(h) Total scheduled maintenance time consumed (in hours).

(2) Unscheduled maintenance. Record the following:

(a) Description of test phase, environment, and meteorological conditions under which maintenance requirements developed.

(b) Procedures and special tools (if any) required to complete maintenance actions; description of maintenance performed.

(c) Identity of, and elapsed time since, previous maintenance (scheduled or unscheduled, whichever is shorter).

(d) Parts required, if any.

(e) Any difficulty in completing maintenance actions.

(f) Components or features involved and method used to determine.

(g) Failure-caused damage to associated parts of the system, if applicable.

(h) Reason for maintenance required, if apparent.

(i) Recommendations to prevent recurring need for maintenance action.

(j) Time required to identify and locate fault.

(k) Time required to correct fault.

(l) Total unscheduled maintenance time consumed (in hours).

(3) Technical engineering investigations or inspections related to analysis of failure cause or detection of suspected incipient failures.

(4) Total maintenance performed. Record the following:

(a) Adequacy of the draft technical manual and maintenance information provided.

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(b) Adequacy of the organizational equipment maintenance tools and repair parts.

(c) Adequacy of the safety instructions.

(5) Total operating time (in hours).

(6) Identify each maintenance action as a chargeable or nonchargeable failure, in accordance with combat and materiel developers' failure definitions and scoring criteria. Indicate total number of chargeable failures and total number of nonchargeable failures.

f. Safety. Record data required by the specific commodity item TOP, AMCR 385-100, AMC Supplement 1 to AR 385-61,[40] FM 31-3,[41] the maintenance package. Additionally, record the following:

(1) Tabular listing of all safety features or devices, including:

(a) Type.

(b) Purpose.

(c) Suitability.

(d) Adequacy.

(e) Method of operation.

(2) Adequacy, clarity, sufficiency, and location of warning plates, instructions, and markings.

(3) Any condition that did or might present a safety hazard, including cause and corrective action required.

(4) Presence and adequacy of fire fighting equipment, and suitability of storage and control locations.

(5) Suggestions to improve safety features, measures, or precautions.

g. Human Factors. Record the following:

(1) Results of appropriate objective tests.

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[40] AMC Supplement 1 to AR 385-61, Safety Program for Chemical Agents and Associated Weapons System, 20 September 1972.

[41] FM 31-3, Weather Support for Field Army Tactical Operations, 4 December 1969.

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(2) Completed subjective questionnaires.

(3) Completed HF error reports.

(4) Completed task/item checklists.

h. Security from Detection. Record the following:

(1) Characteristics of terrain in which observations were made.

(2) Equipment used to make observations (e.g., field glasses, aircraft).

(3) Distance from observer to target.

(4) Time of day and sky conditions when observations were made.

(5) Type of materials or materiel used.

i. Value Analysis. Record comments for topics listed below:

(1) Mission capacity.

(2) Simplicity.

(3) Level of technology.

(4) Standardization.

(5) Materials and methods of construction.

(6) Tolerances.

j. When making recommendations for changes in item features or components, record comments on: (a) the feature or component under consideration, (b) recommended change, and (c) reason for recommended change.

6.2 Analysis

a. General. Summarize all data using tabular listings, graphs, or charts, as appropriate. Analyze and compare data collected against specific criteria in requirement documents. Use data from control group (items stored in air-conditioned area not influenced by tropic effects) as basis for stating presence of tropic environmental effect. Provide a narrative description of the degree of suitability of the test item for use in the tropics. If the item is determined not suitable for use in the tropics, provide reasons (including test results) for that determination.

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b. Questionnaires. Interview personnel, using structured questionnaire/checklists, asking for opinions on specified features in a particular test.

c. Summarize questionnaires/interviews/checklists, and include in final test report, as appropriate (TECOM Pamphlet 602-1[42]). Group the answers to all checklist questions under headings (e.g., safety, accessibility, human factors) and arrange answers in order of relative importance.

d. Logistics Supportability. For information on meaning and analysis of maintenance data, see TECOM Supplement 1 to DARCOM Regulation 700-15.[43]

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[42] TECOM Pamphlet 602-1, Volume 1, Questionnaire and Interview Design (Subjective Testing Techniques), 25 July 1975.

[43] TECOM Supplement 1 to DARCOM Regulation 700-15, Integrated Logistics Support (ILS), 20 June 1980.

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